EZSCAN

User guide

Instruction for use

Version 2.3

To download the latest version of the software and the manual please go to
www.impeto-medical.com/distributor-resources



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Introduction

Use

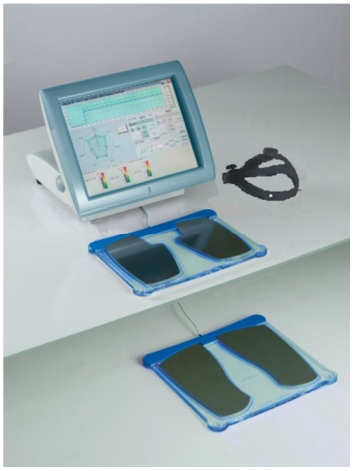
EZSCAN is a medical device for the **early detection of Type 2 diabetes risk and complications**.

Principle

EZSCAN measures the effect of both early and established diabetes on the eccrine sweat gland system. Changes in the ionic balance (Cl^-, H^+) of the sweat ducts can be used to assess the patient condition in relation to early development of Insulin Resistance and Type 2 diabetes.

Description

EZSCAN is an electro-medical device that consists of:



- **A control panel and display**
- **A pair of foot electrodes**
- **A pair of hand electrodes**
- **A pair of head electrodes**
- **A power cable**

EZSCAN is powered by Windows XP and/or Windows7..

Installation

Carefully follow instructions in this manual to ensure correct installation.

Complete installation should take no more than 15-20 minutes.

1. Unpacking

EZSCAN is shipped in a single transportation box with an Impeto Medical logo on the outside. Inside there are 3 smaller boxes:

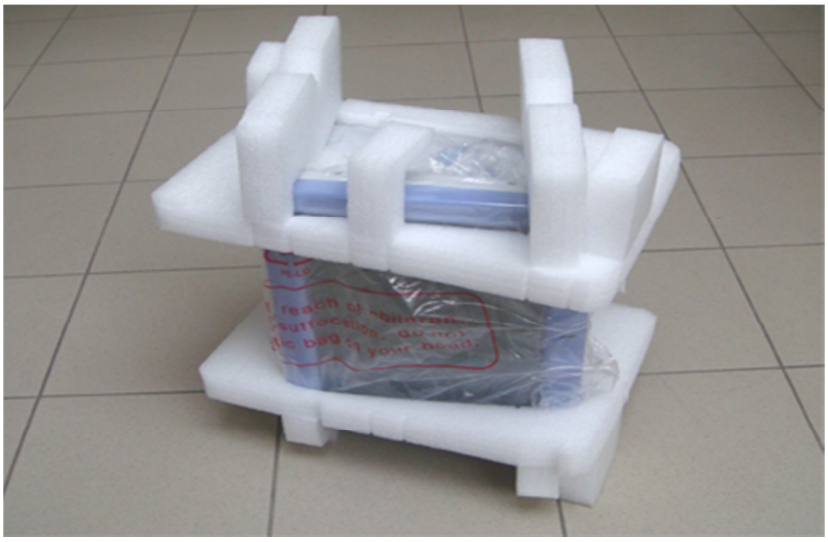
1. A medium white box containing the reusable feet and hand electrodes,
2. A small white box containing the auxiliary cable and the reusable head electrodes,
3. A big white box, with the EZSCAN® Master Unit and the power cable.



1.1. Unpack the master unit

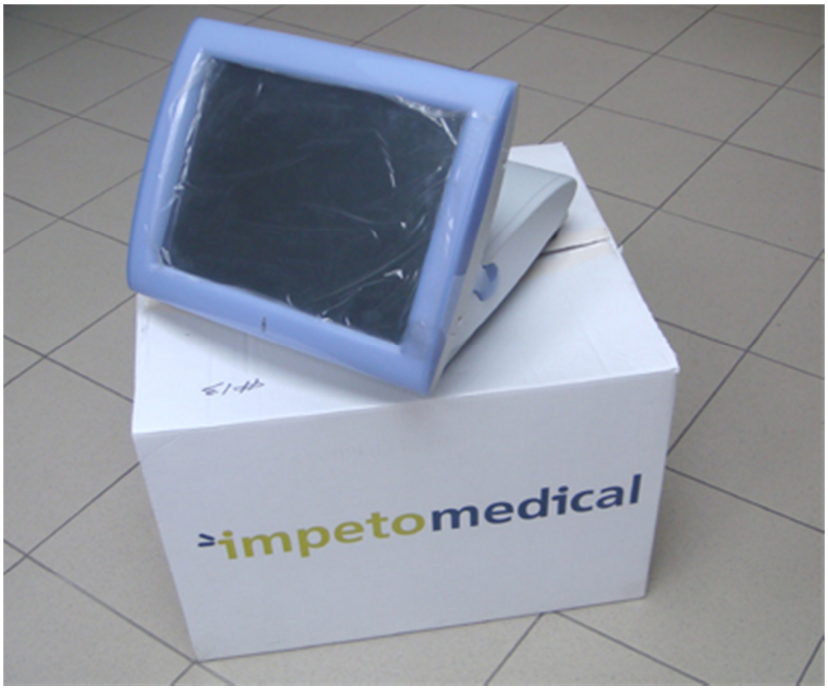
1	Remove the Master Unit along with its 2 molded protection foams from the big white box.
2	Put the Master Unit on one side to remove the first protection easily and safely the first protection foam.
3	Put the Master Unit on its other side and remove the second protection foam, as well as the plastic bag.

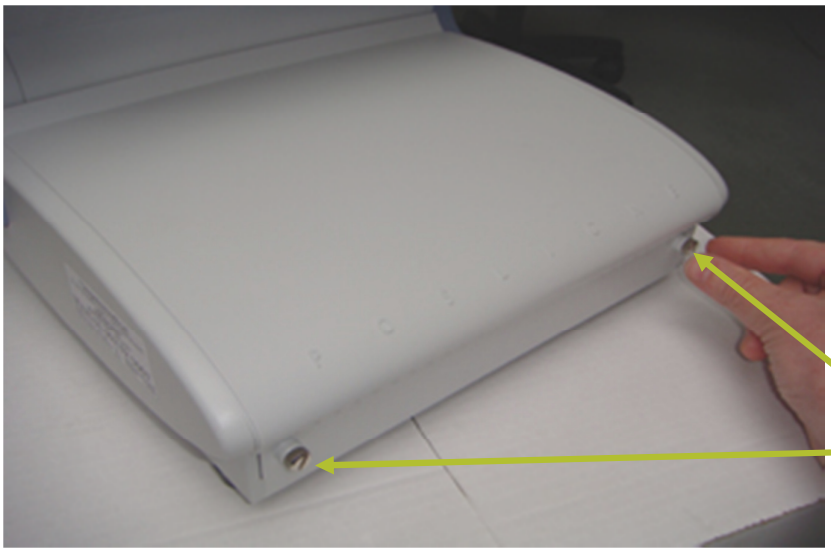


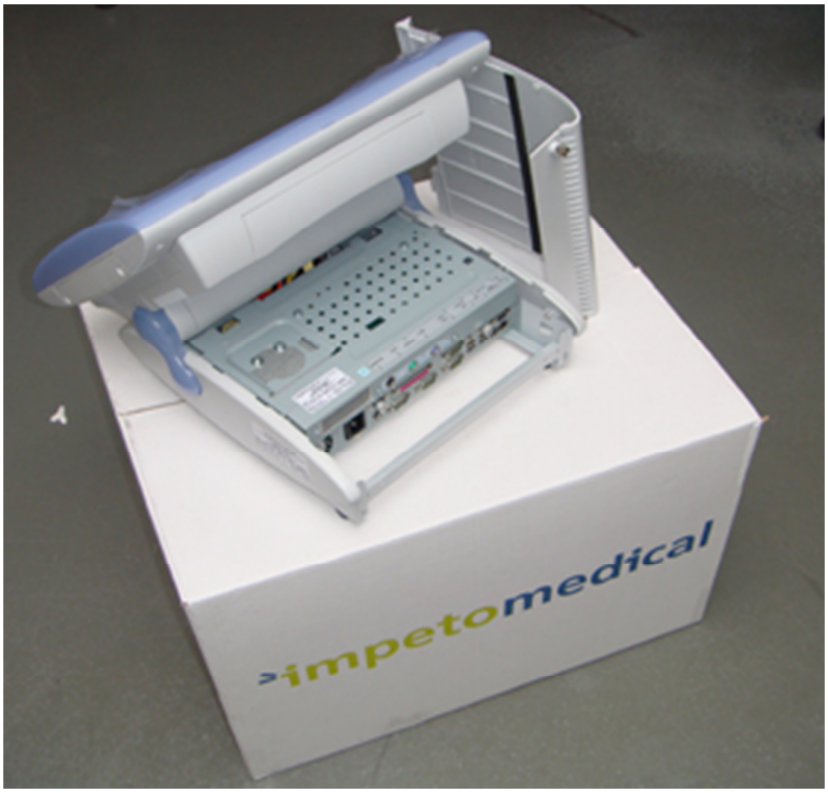


4 Keep the protection foams and plastic bag in the outer box, for future transportation needs.

5 Loosen the 2 retaining screws and remove the rear cover.


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Retaining
screws


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1.2. Unpack the electrodes

1 Remove the electrodes from the medium white box and separate them from their protective foams.



2 Remove the head electrodes cable and the cleaning spray from the small box.

3 Remove all 3 cables and connectors from their plastic protection bags.


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2. Installation

1 Place the Master Unit on a desk or preferably a high table (h > 90 cm/3ft) with the screen facing the user.



NOTE

Please consult your distributor for the availability of an optional EZSCAN companion rolling stand which can also conveniently host a printer.

Connecting the electrodes

2 Place the hands electrodes in front of the Master Unit, guide the cable under the metallic bar at the rear, and insert the MiniDin connector at the end of the Hands cable to the 'HANDS' connector (Green) at the back of the Master Unit :



3 Make sure to match the color of both socket and plug.

4 Push until a ‘click’ sound indicates the locking of the connector.



5 Double-check the locking by grabbing the cable itself gently backwards.



6 Place the Feet electrode on the floor in front of the Master Unit and connect its cable to the 'FEET' connector (Blue) at the back of the Master Unit, using the procedure above.

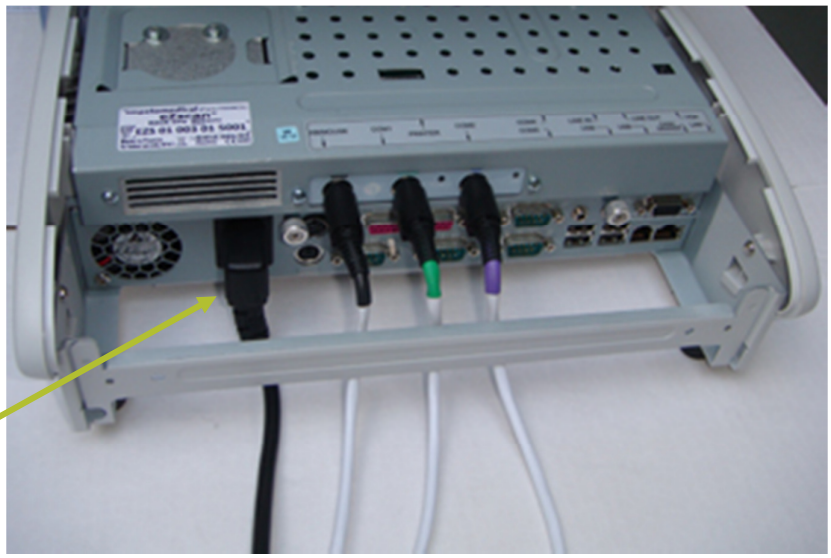
7 Connect as well the Head electrodes cable to the 'HEAD' connector at the back of the Master Unit using the same procedure.

WARNING

Please cross check and verify that all electrode cables are connected to their respective receptacle (Head to Head, Hands to HANDS and Feet to FEET) and there is no inadvertent swapping.



8 Guide the power cable under the metallic bar at the rear of the Master Unit and insert it into the PC standard 3-plug power socket at the back of the Master Unit.



9 Connect the power cable to the mains power socket on the wall. Double check the power socket can provide proper grounding.

NOTE

The supplied power cable is intended for EU countries.

For countries outside EU, please make sure using a national PC-type power cable with grounding connector. Make sure also correct grounding is provided at the wall socket.

3. Quick test and configuration

3.1. Powering on EZSCAN

1 Open the door located below the screen at the front of the Master Unit.
Inside on the right there are 2 USB ports and 1 on/off button.



2 Press and release the on/off button to power up the Unit.
The Master Unit will run a self-check and then automatically display the EZSCAN home page.



3.2. Initial checking

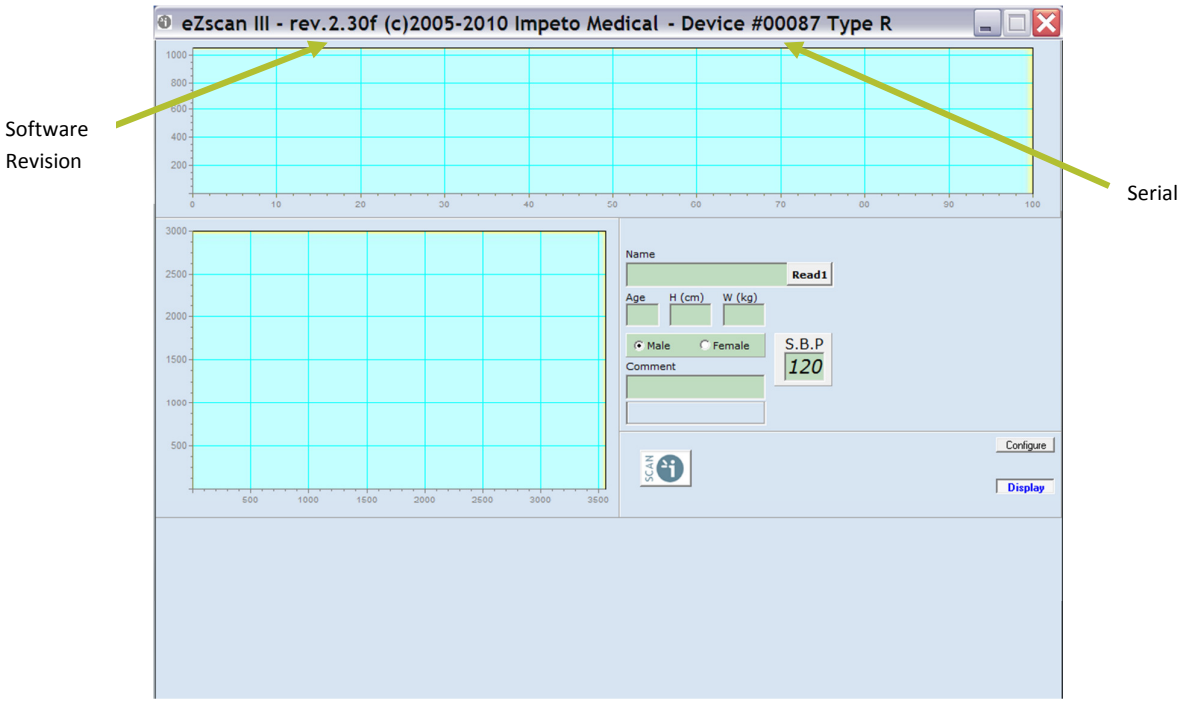


After initialization, the blue LED at the rear of the Master Unit should flash intermittently. The top row on the screen should display:

eZscan - rev. X.XXx ©2005-2010 Impeto Medical - device #YYYYY type Z

Where:

- X.XXx indicates the software revision number,
- YYYYY indicates the serial number within the product model type Z.



IF: 1/ the blue LED is NOT ON

OR: 2/ the following message appear at the top of the screen:

"eZscan - rev. X.XXx ©2005-2010 Impeto Medical – not connected",

There is a problem with your EZSCAN Master Unit. Please call Impeto Medical or your distributor for assistance.

Once this quick check procedure is successfully completed, replace the rear cover at the back of the Master Unit. Secure the cover in place using the 2 retaining screws.



3.3. Selection on the EZSCAN tactile screen

EZSCAN has a tactile screen. Touch the screen with your finger to make a selection.

You can also use a mouse by plugging one into one of the USB port.

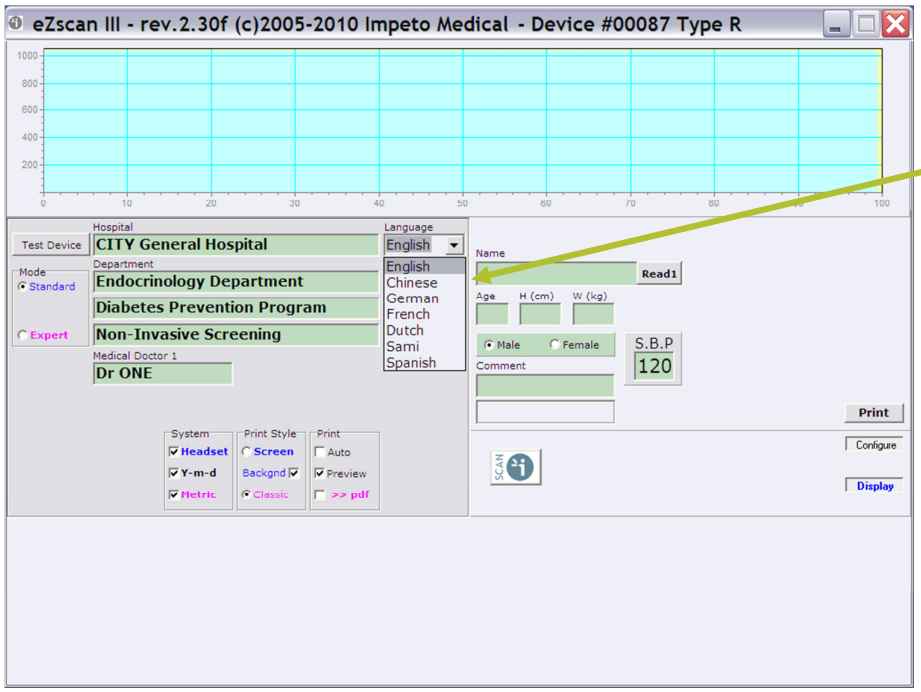


3.4. Language selection

EZSCAN software can be used in several languages:

- English
- Chinese
- German
- French
- Dutch
- Spanish
- …

Select one of the languages in the language menu.



Language menu

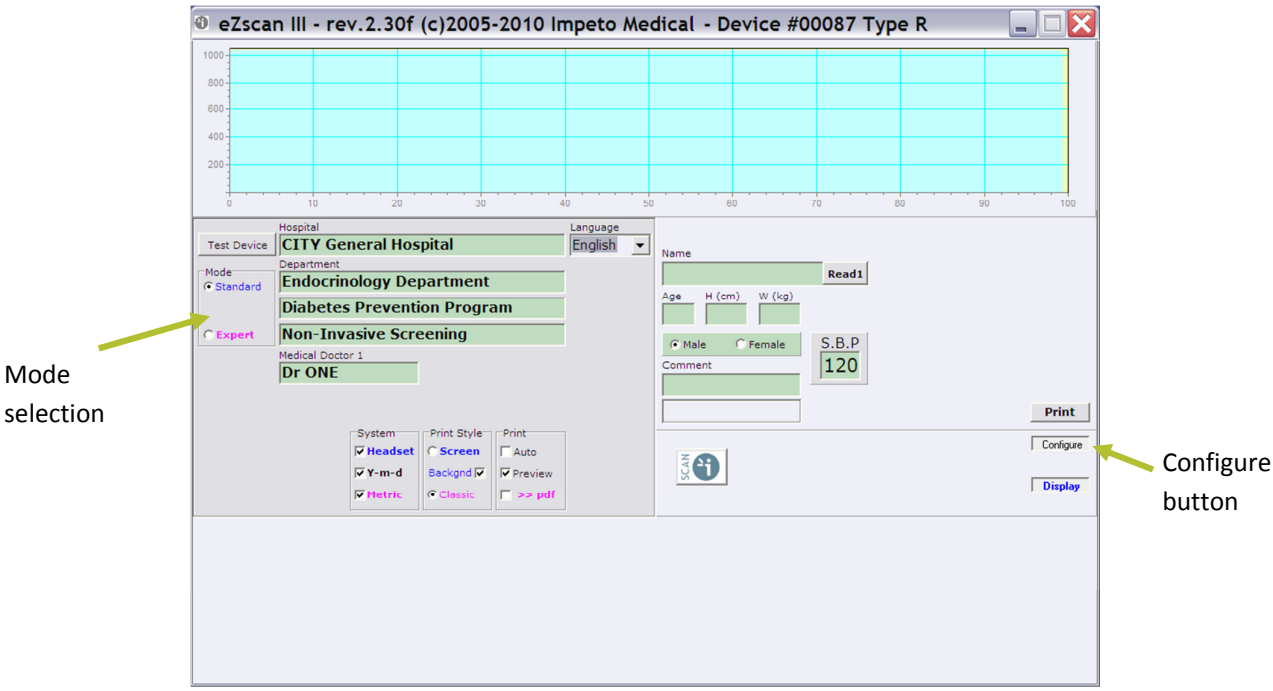
3.5. Mode selection

Two modes are available on EZSCAN:

The standard mode is the preferred mode to easily scan a new patient. It is a simple mode showing only the information needed to screen new patients.

The expert mode is the preferred mode to compare with scans previously done.

Check the mode button according to your choice.



NOTE

Pressing the “Configure” button on the right will close/open the Configuration window.

3.6. Metric selection

Two measurement systems are available for patients height and weight:

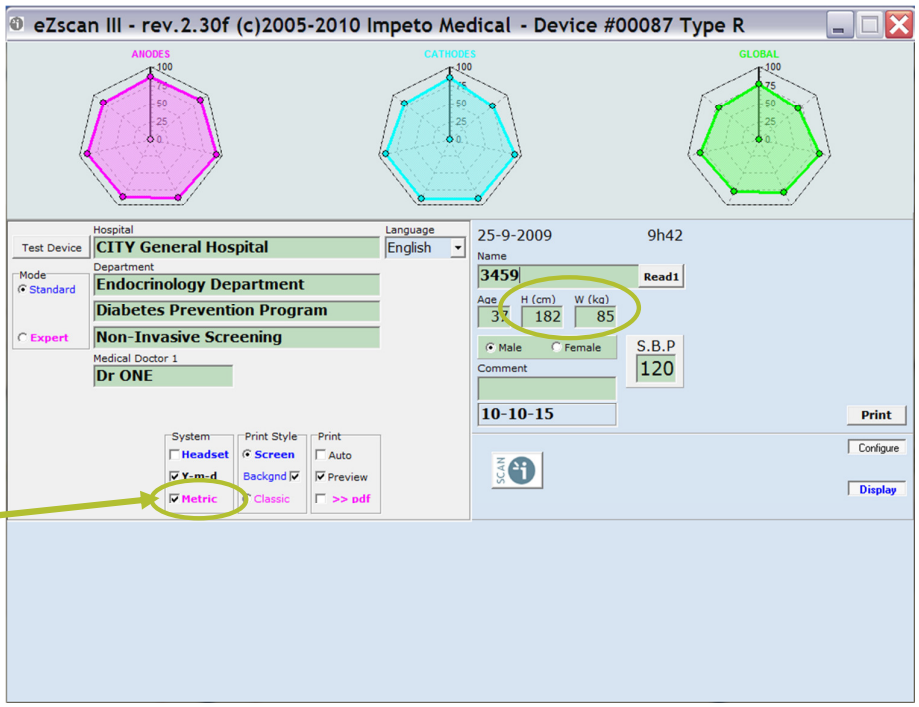
- Meter (cm) and kilogram (kg)
- Feet/inch (feet/in) and Pound (lb)

Check the Metric box if you wish to use cm and kg.

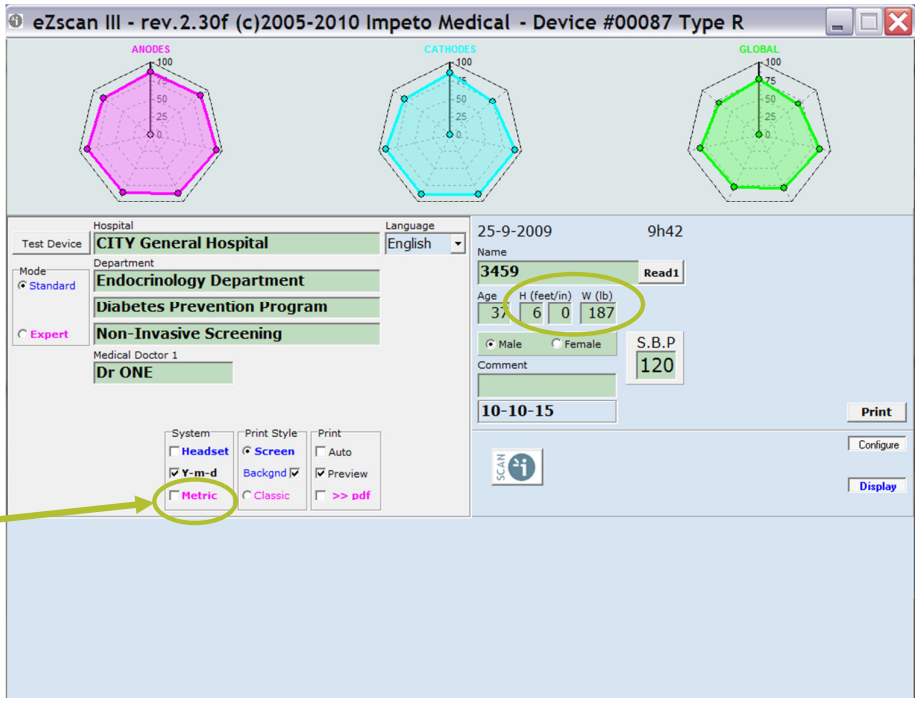
Uncheck the Metric box if you wish to use Feet and Id.

The measurement will be instantly modified.

Metric checkbox



Metric checkbox



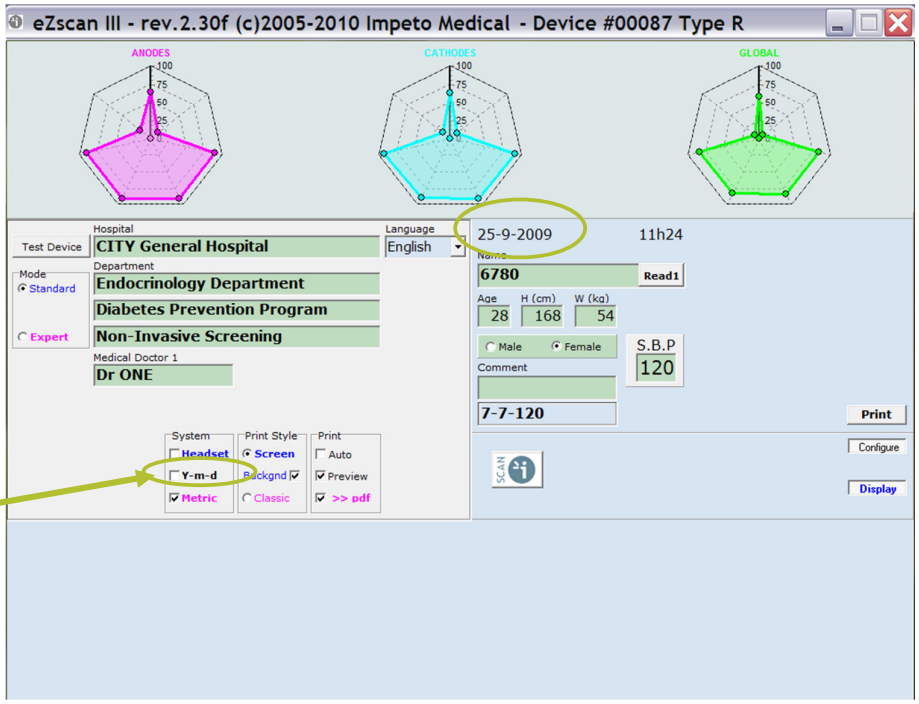
3.7. Date format selection

Two date format are available: Day Month Year (D-m-Y) or Year Month Day (Y-m-d)

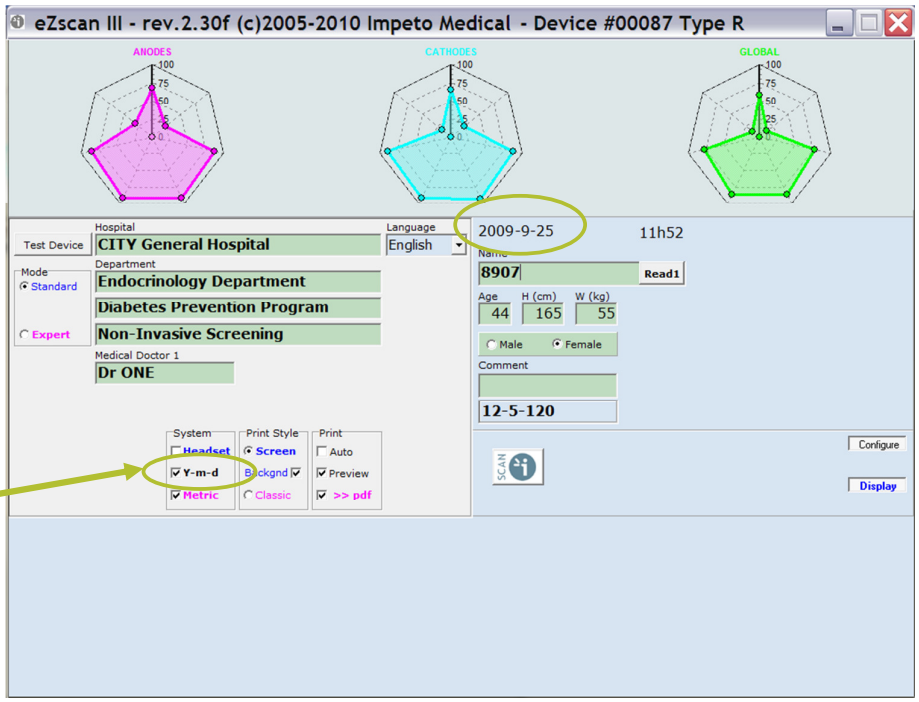
Check the date format checkbox if you wish to use Y-m-d format.

Uncheck the date format checkbox if you wish to use D-m-Y format.

This option will be taken into account for the next scan.



Date format checkbox

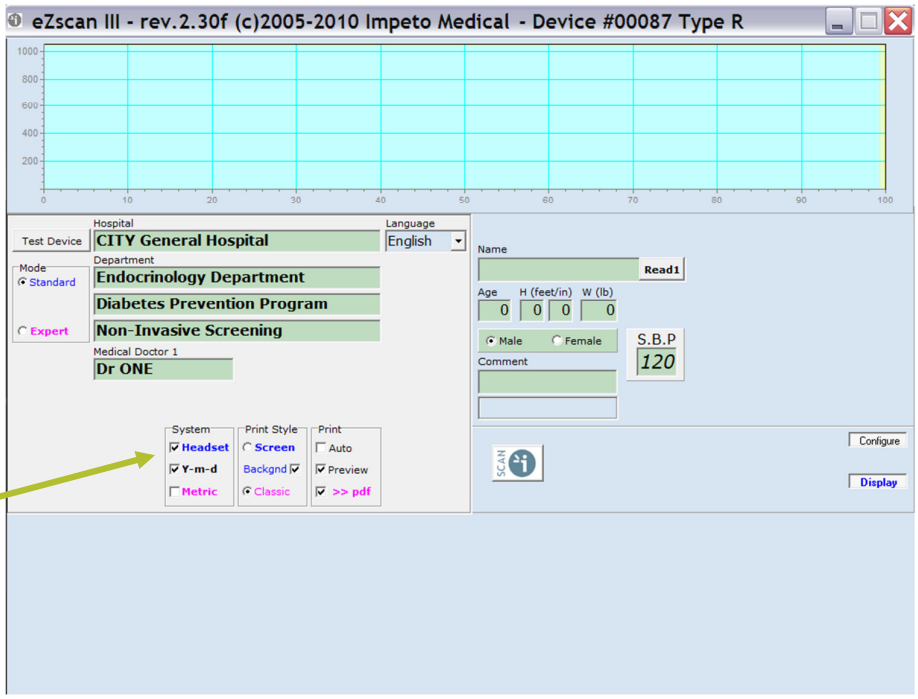


Date format checkbox

3.8. Headset selection

In order to increase the patient throughput on eZscan device, and the speed of screening large populations, it is possible to remove the Headset electrodes, which will reduce drastically the patient installation time as well as the scanning time.

Uncheck the Headset bow if you wish to use EZSCAN without the head electrodes.



Headset checkbox

NOTES

Only when the screening strategy is selected (cf. section 5.), the head electrodes are optional : follow-up strategy requires the headset electrodes.

Without headset, the Strategy Mode will be automatically forced and limited to screening only.

After deciding not to use a Headset electrodes, please double check this option is unchecked, in case you experience a “Please check Headset…” error message at the beginning of a new scan.

4. Standard Mode

The standard mode is used to easily scan a new patient. More options are available in the Expert mode (cf. section5.). Check the options selected in the Expert mode before starting a run.

4.1. Quick run

Before starting a scan, you will need the patient to stand barefoot on the Feet electrodes and input a few patient details (name, age, height, weight, gender) using the touch screen display and the virtual keyboard appearing on the screen.

Inform the patient that this test is non-invasive and will take less than 3 minutes to perform.

NOTE

For conventional data input, it is also possible to use any standard PC-type keyboard connected to one of the 6 USB connectors available on Master Unit's front or rear panels.

4.1.1. Patient set-up

1	Ask the patient to remove their shoes and socks.
2	Patient must stand, with bare feet right in the middle of the Feet electrodes.
3	Put in place the head gear with reusable electrodes and adjust its size to the patient head.
4	Connect the blue side of the cable to the electrode placed on the right side of the patient's head and the white side on the left. *Make sure there is no hair between the skin and the electrodes.*



5

Input patient data using the touch screen virtual keyboard.

Enter each data field by firmly double-clicking on the corresponding green zones, which will make the appropriate virtual keyboard appear.
The TAB key on the keyboard can also be used to jump from one field to the next.

Please key in the correct information for each new patient:

- *NAME or Patient ID (IDENTITY CODE)*
- *AGE*
- *HEIGHT*
- *WEIGHT*
- *Select the appropriate GENDER*
- *Enter comments in the "comment" field if necessary.*

There is NO NEED to press the ENTER key after each field entry.

NOTE

Double-check the data with your patient. An invalid data will be notified with a RED background and should prompt entering a valid data.

5

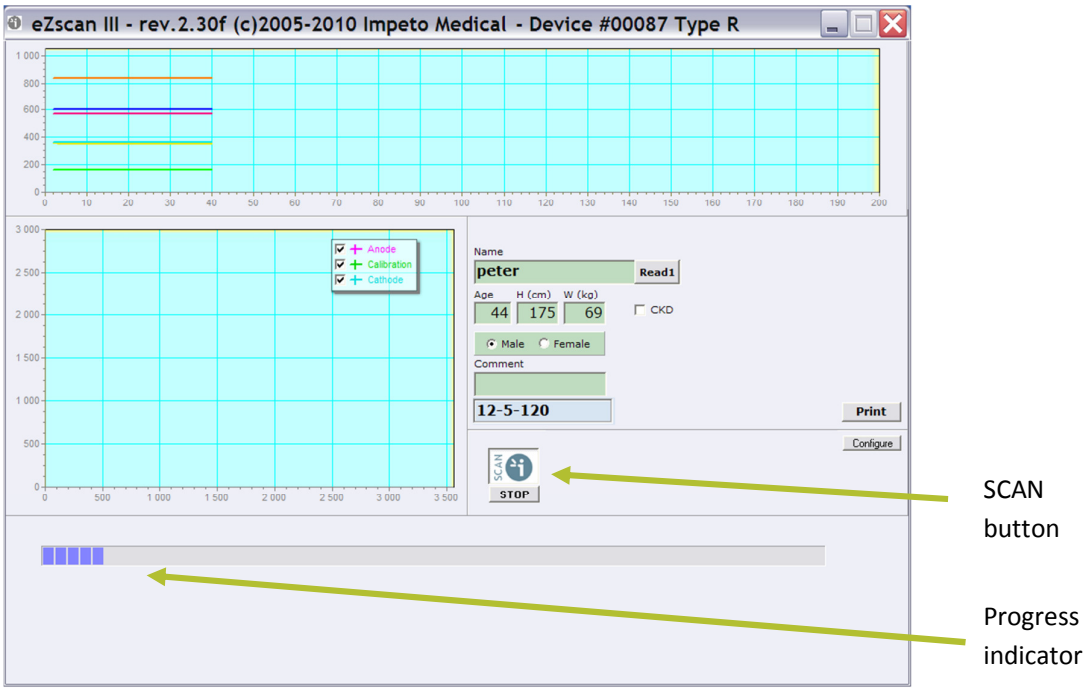
Ask the patient to put his hands flat in the middle of the Hands electrodes.
Hands must be applied firmly.



4.1.2. Start the measurement

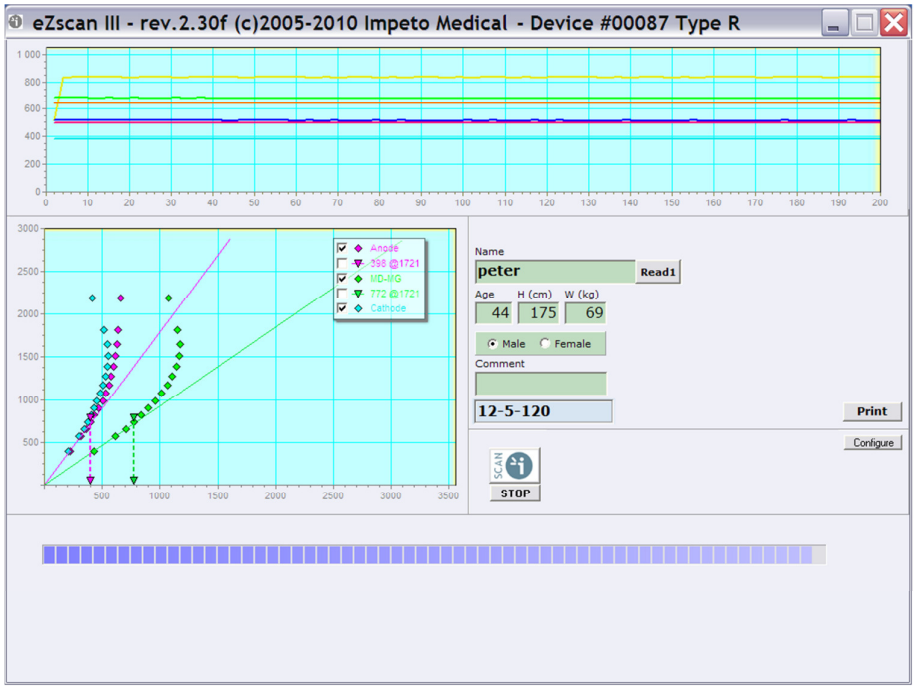
Press the “SCAN” button to start the scanning procedure.

During scanning, the patient is required not to move any parts of the hands or feet in contact with electrode surfaces.





The **patient should keep still** until the progress indicator shows the scan is completed.



Scan progress can be monitored from the indicator at the bottom of the display.

Scan completion is confirmed by a double-bell sound.

4.1.3. Save patient data

Upon completion of the scan, results are automatically displayed and all patient's data are automatically saved onto the Master Unit's hard disk.

4.2. Clean the electrodes

IMPORTANT

Immediately after each scan, it is important to clean all electrodes with the proper cleaning spray and soft tissues*.

It is not only necessary for health reasons, but it will also 'neutralize' the electrochemical reactions which have taken place on the electrodes during the scan, stopping the corrosion process which would otherwise quickly damage the electrodes surface.

1. Deposit only a small quantity of cleaning product on each electrode.

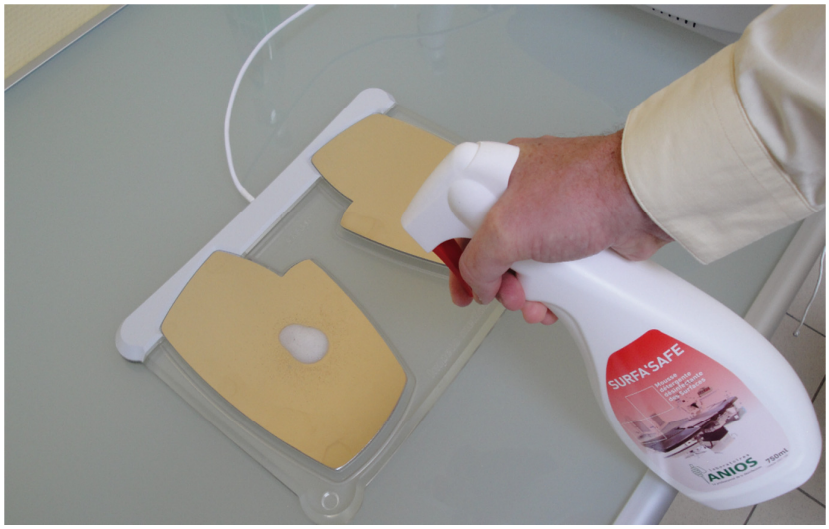
   Avoid applying excessive cleaning liquid.

2. Wipe electrodes with a sheet of soft paper towel to until dry.

   Use a new sheet for each electrode or as long as the surface of the electrodes are not completely dry.

HINT

Take advantage of the shinny mirror aspect of the electrodes to remove all traces from all areas!



***Impeto Medical recommends using Surfa'Safe from ANIOS Laboratories (France) or equivalent.**

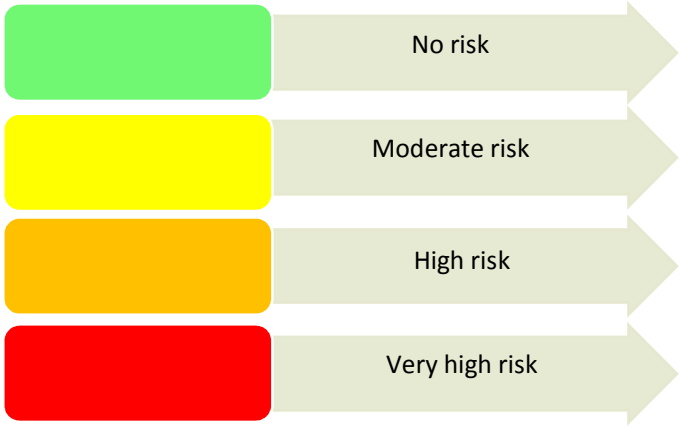
One 750ml bottle will allow to clean all electrodes after each of 750 patients.

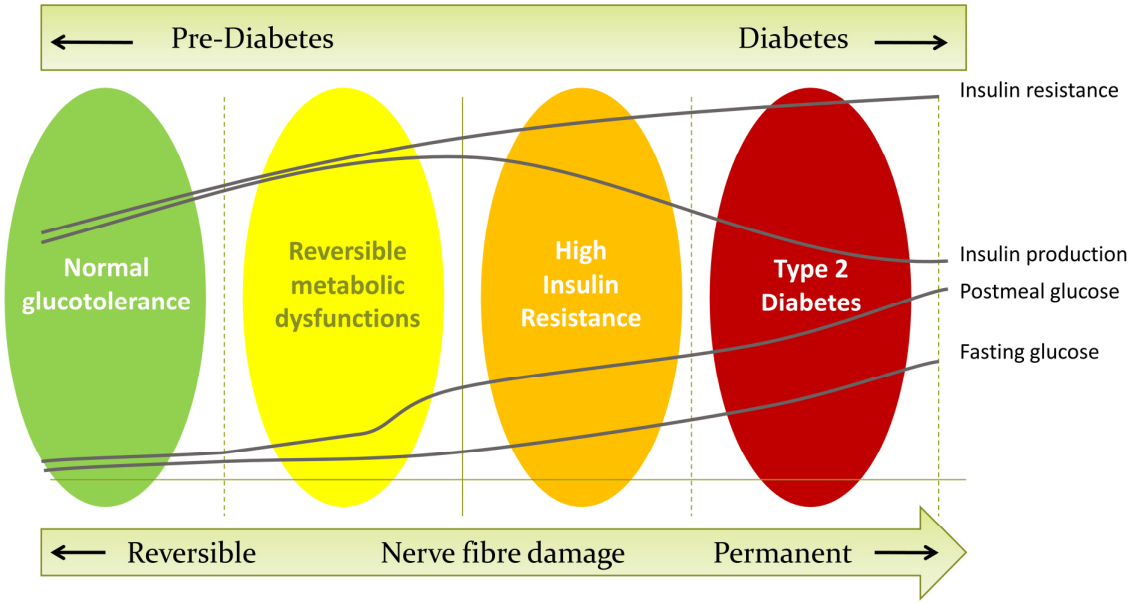
4.3. The results

Results should be interpreted by a medical professional and take into account the patient's clinical history.

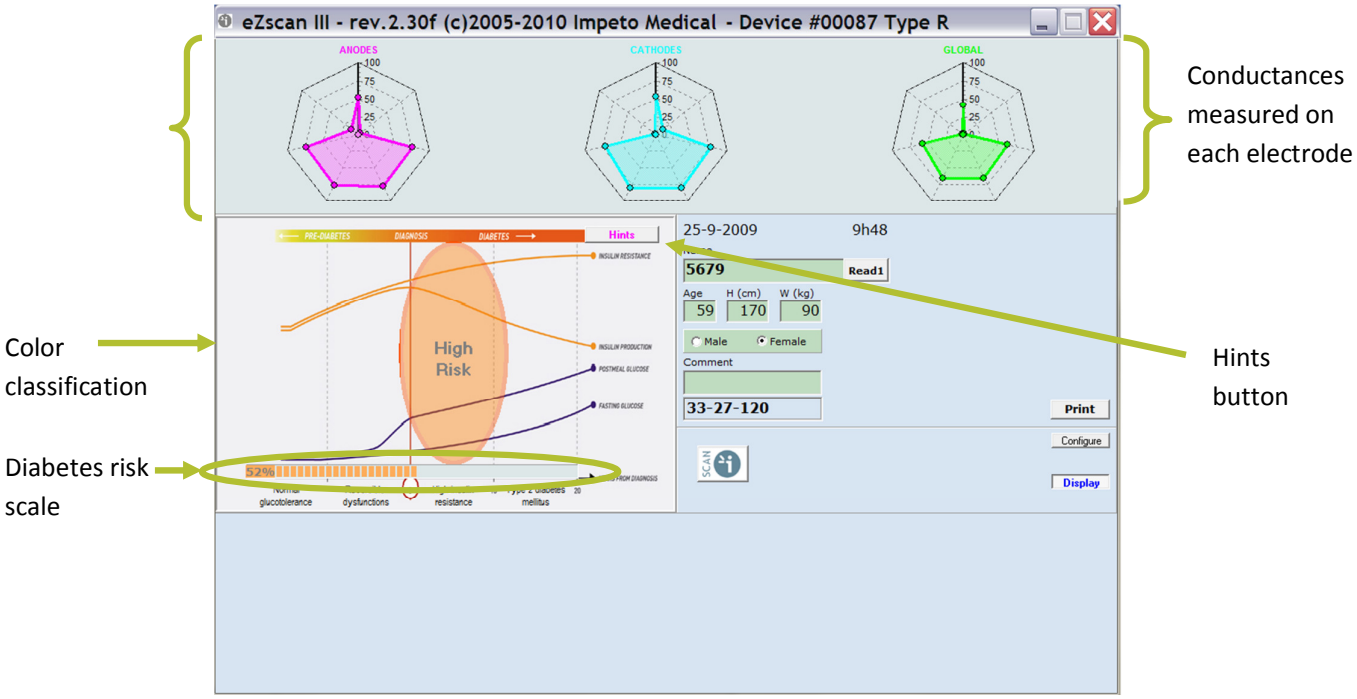
4.3.1. The screen results

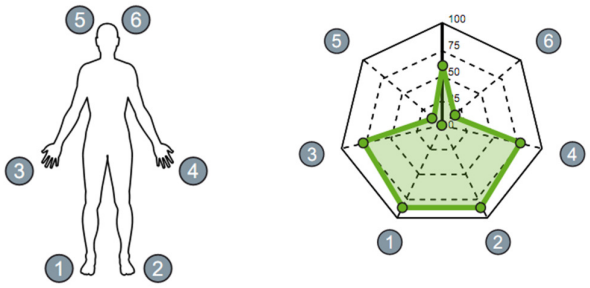
Following a three minute data sampling period, EZSCAN allows simple reading of results with the colored-ellipse positions which indicate where the patient is on the diabetes historical curves'. Results are displayed according to diabetes risk with a colour index.



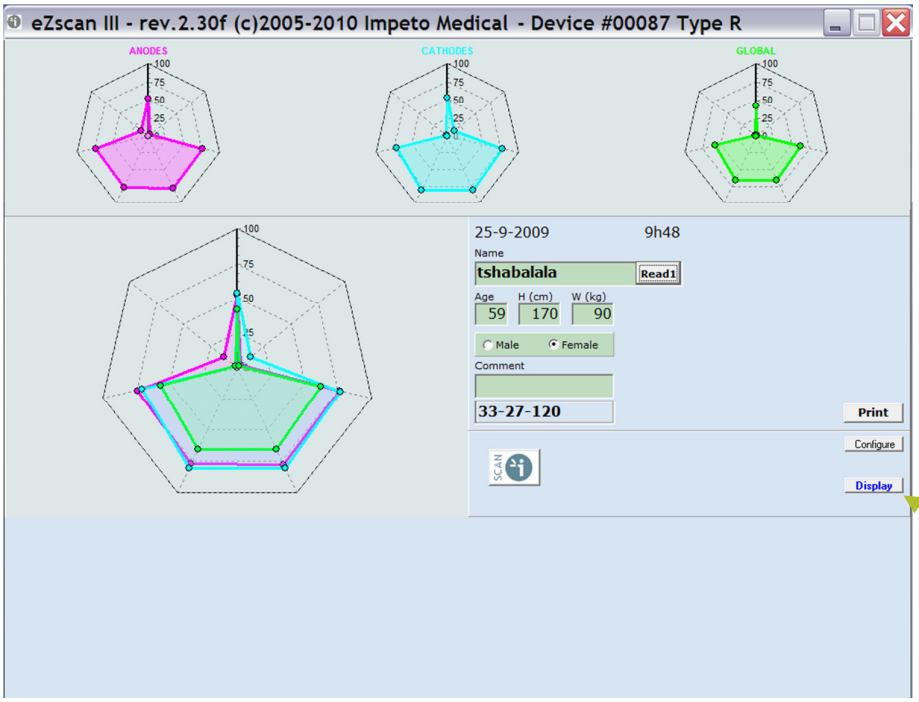


Three radars on the top of the screen show the electro conductances measured on each electrode. A diabetes risk scale is displayed under the color classification.





The superposition of the conductance graphs can be obtained by clicking on the Display button :

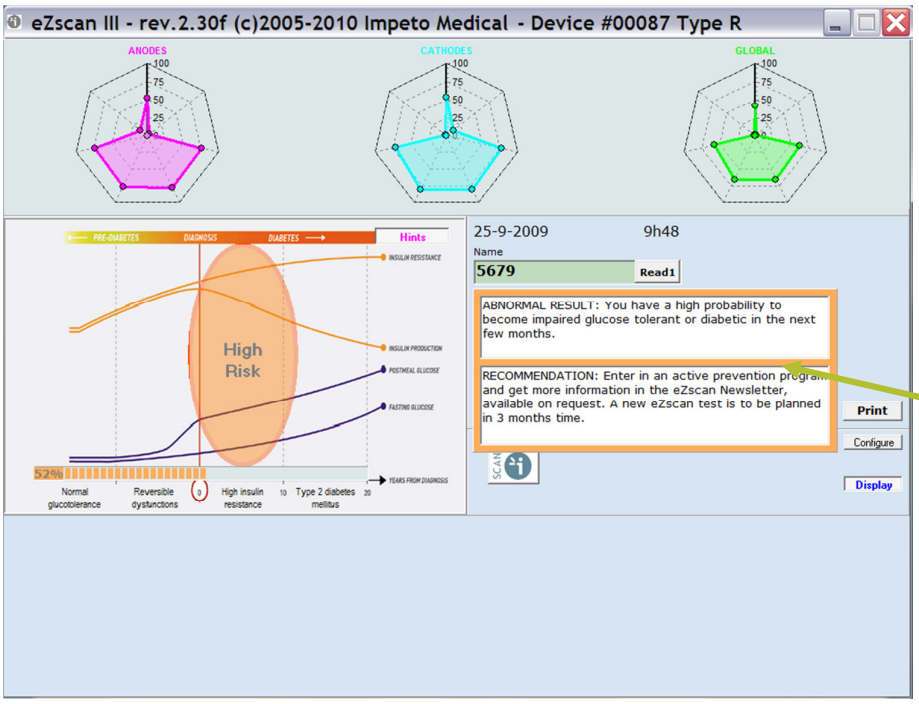


Display button

4.3.2. Quick hints

According to the subject color classification, some hints are given to the subject.

Click on the Hints button to display the hints.

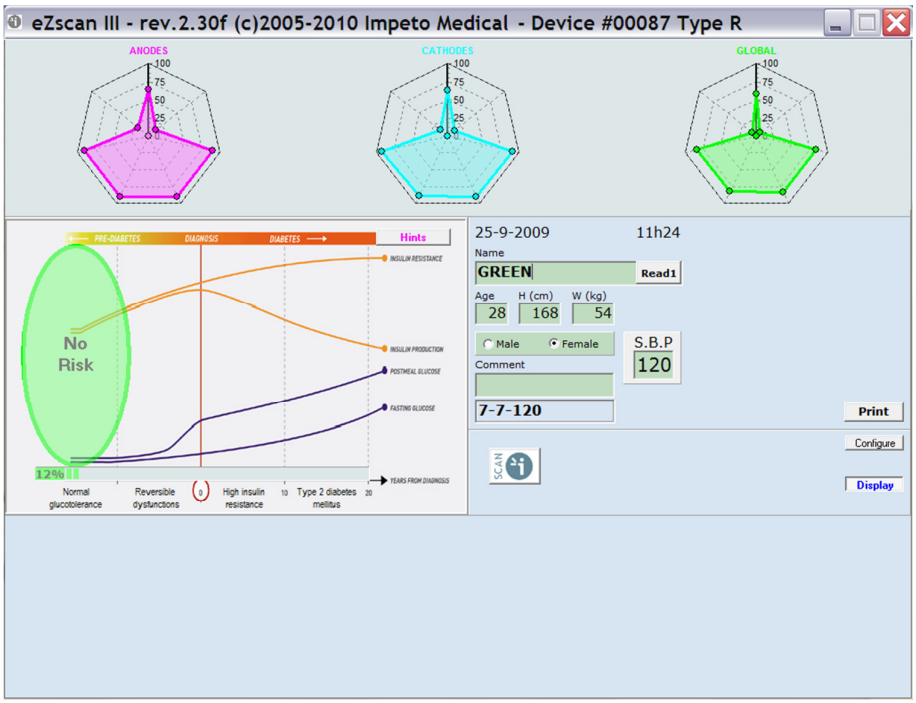


The hints can be modified:

1. Double click on the Hints text zone. *The keypad appears at the bottom of the screen.*
2. Modify the text.

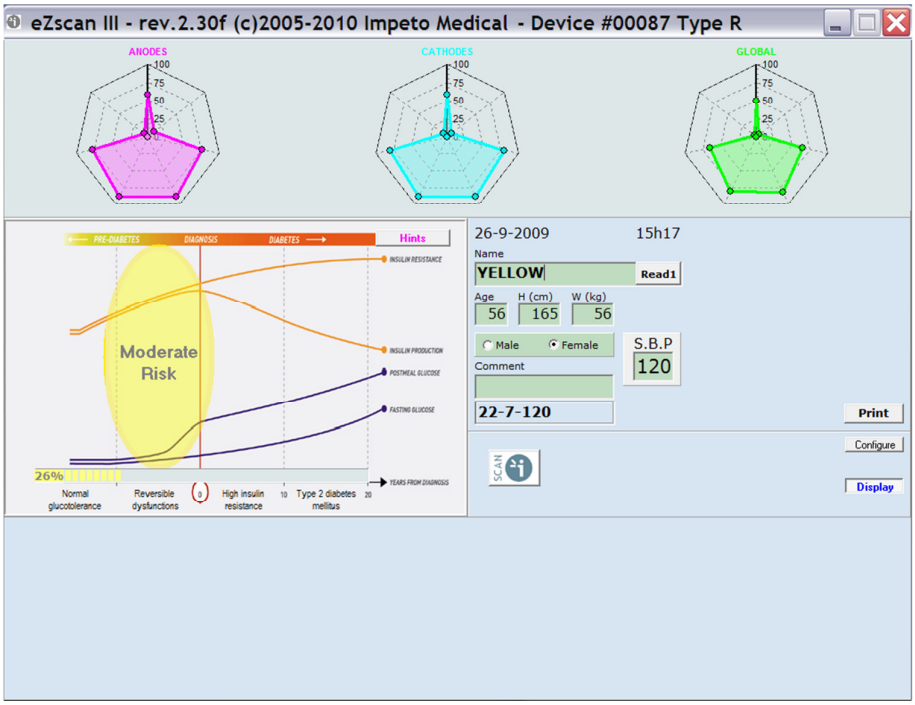
4.3.3. Quick interpretation of the results

4.3.3.1. Green patient



The green result corresponds to NGT (Normal Glucose Tolerance). The patient has a high probability of not having an abnormality in his glucose metabolism and with this, not an increased diabetes risk.

4.3.3.2. Yellow patient

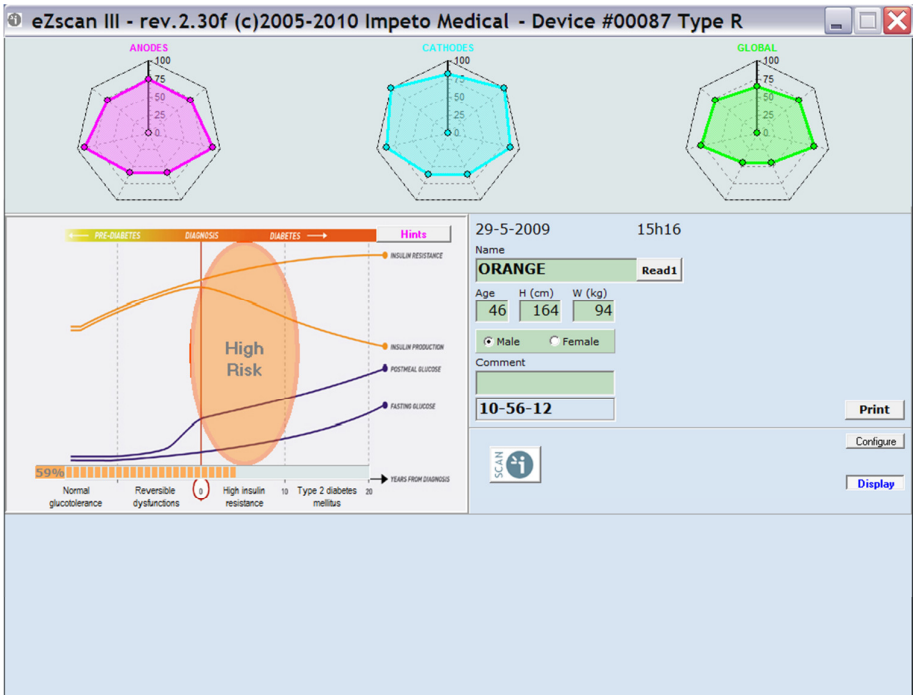


The process of nerve damage has already started but still unlikely to be detected by FPG (Fasting Plasma Glucose) Test. IGT can usually be found by an OGTT (Oral Glucose Tolerance Test)…which requires 2 blood sample tests separated by 2 hours!

The patient has a low probability of having an abnormality in his glucose metabolism but an increased risk in getting diabetes in the next years. At this stage, just paying more attention to nutrition habits and increasing exercise can reverse the disease process. Results can be seen after few weeks with a new EZSCAN test.

Regular tests are required to monitor patient's condition and prevent progression to T2DM (Type 2 Diabetes Mellitus).

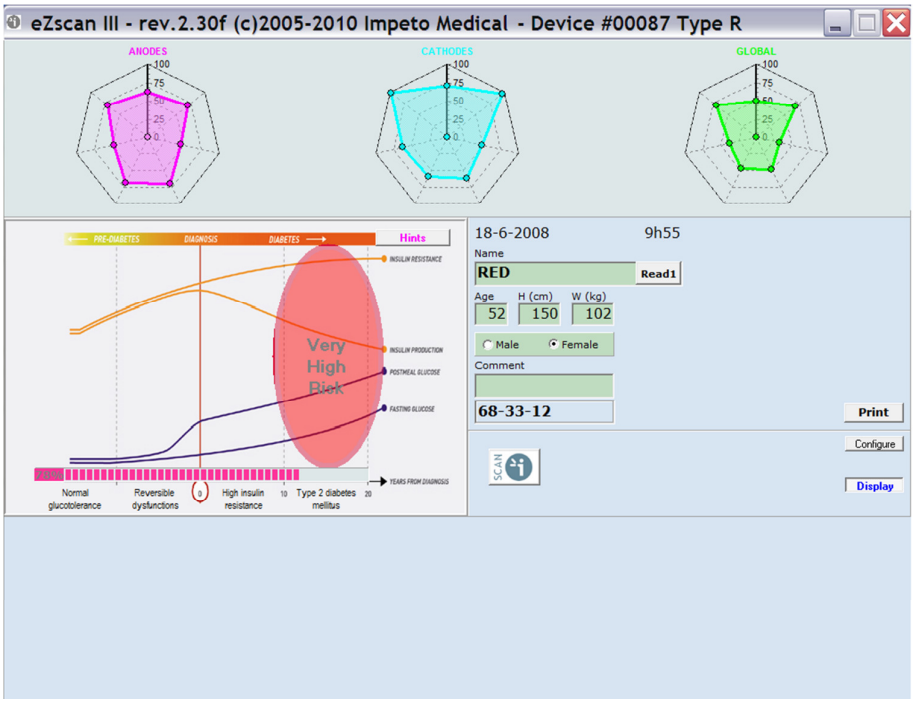
4.3.3.3. Orange patient



Metabolic disorders are present. The patient has a high probability to become impaired glucose tolerant or diabetic in the next few months.

Insulin resistance can lead to T2DM which is the beginning of measurable gap between insulin need and insulin production with small side effects. There are usually no physical symptoms perceived by patients until a few years later. Patient should be tested for metabolic status with blood pressure, OGTT, Cholesterol and triglyceride levels and will have to be checked every 3 months to follow up.

4.3.3.4. Red patient



The results are clearly abnormal! T2DM (Type 2 Diabetes Mellitus) seems already in place. Patient should be referred for OGTT and HbA1c. Severe complications become possible.
Regular EZSCAN tests may help in assessing emerging autonomous neuropathy (cf. Expert mode in 5.), which might be useful to prevent the patient developing complications like:

- Peripheral neuropathy with possible foot amputation,
- Cardiac neuropathy with increased risk of heart attack,
- Erectile dysfunction,
- Renal impairment,
- Retinopathy,
- Etc…

5. Expert Mode

Proceed as described in 4. to run a test.

The expert mode can be used in the context of a follow-up scan. Two strategies are available: Screening and Follow-up.

5.1. Compare two sets of results

In both strategies, it is possible to compare two results. This function can be useful during a follow up scan. It enables the doctor to assess patient progress.

1. Press the 'READ 2' button to open the current folder within the currently selected Medical Doctor.

   To select another Medical Doctor (MD1, MD2, etc), click on the 'up' button  and select the desired MD.

2. Double-Click on the patient's name you want to retrieve the data.

NOTE

You can switch from patient 1 data to patient 2 data, by simply touching the corresponding name field and the diagnosis will be updated accordingly for each.